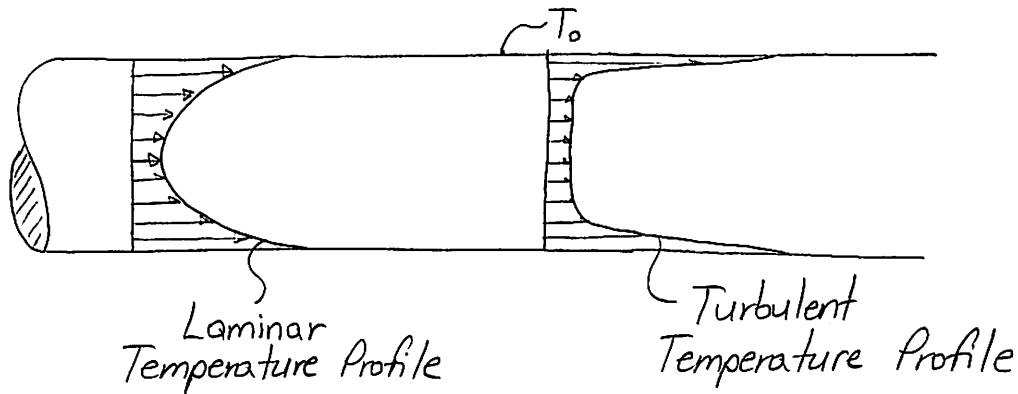
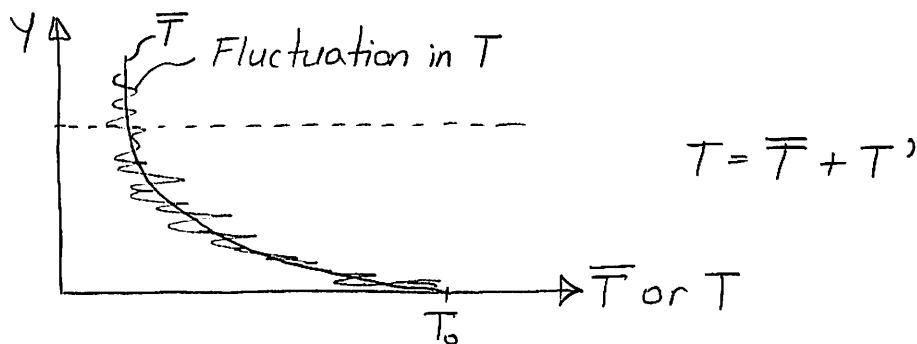


Note however we want to solve for heat transfer



We can follow the exact same procedure as we did with the momentum balance, but now with energy



$$\bar{q}'' = -k \frac{\partial \bar{T}}{\partial y} + \overline{v'T} \rho C_p \Rightarrow \text{Energy integral formulation}$$

$$\bar{q}'' = -k \frac{\partial \bar{T}}{\partial y} + \overline{v'T'} \cdot \rho C_p$$

$$\overline{v'T'} = -\epsilon_T \frac{\partial \bar{T}}{\partial y}$$

↓ Turbulent Prandtl Number

$$Pr_t = \frac{\epsilon}{\epsilon_t}$$

Now we can do the following:
So our heat transfer becomes:

$$\boxed{\begin{aligned} \bar{q}'' &= -\rho C_p (\alpha + \epsilon_T) \frac{\partial \bar{T}}{\partial y} \\ \bar{T} &= \rho (\nu + \epsilon) \frac{\partial \bar{u}}{\partial y} \end{aligned}}$$

We can now make a useful analogy
(Turbulent Colburn analogy)

$$\frac{\overline{q''}/\Delta T}{\rho C_p U} = \frac{1}{2} \left(\frac{\overline{C}_o}{\frac{1}{2} \rho U^2} \right) \frac{1}{Pr^{2/3}} = \frac{C_f}{2 Pr^{2/3}} = \frac{f}{8 Pr^{2/3}}$$

$$St = \frac{C_f}{2} \cdot \frac{1}{Pr^{2/3}} = \frac{f}{8} \cdot \frac{1}{Pr^{2/3}} \Rightarrow \text{Stanton number}$$

$$St = \frac{Nu_D}{Re_D \cdot Pr} = \frac{\overline{h}}{\rho C_p U} \Rightarrow \text{Note most results are plotted as } St \text{ vs } Re_D, \text{ for experiments.}$$

Aside: Comes from:

$$\frac{Nu_D}{Re_D \cdot Pr} = \frac{\frac{\overline{h} D}{k}}{\left(\frac{\rho U D}{\mu} \right) \left(\frac{\rho C_p}{k} \right)} = \frac{\overline{h}}{\rho C_p U} = \frac{\overline{q''}/\Delta T}{\rho C_p U}$$

Note, this is the same as we had before in laminar flow over a flat plate called the Colburn analogy.
See page 90 of the class notes.

$$St \cdot Pr^{2/3} = \frac{C_f}{2} = \frac{f}{8} \quad (\text{For smooth pipes})$$

$$\frac{Nu_D}{Re_D \cdot Pr} \cdot Pr^{2/3} = \frac{f}{8} \Rightarrow \text{from page 149, } f = 0.184 Re_D^{-0.2}$$

$$\frac{Nu_D}{Re_D \cdot Pr^{1/3}} = \frac{0.184}{8} Re_D^{-0.2}$$

$$Nu_D = 0.023 Re^{0.8} Pr^{1/3} \Rightarrow \text{Turbulent pipe flow Nusselt \# Analytical Result}$$

Valid for: $4 \times 10^4 \leq Re_D \leq 10^5$

Note, the experimental result is:

$$Nu_{D,exp} = 0.023 Re^{0.8} Pr^{0.4} \Rightarrow \text{Dittus-Boelter equation}$$

$\Rightarrow Re_D > 10,000$

Most accurate correlation to use is Gnielinski correlation for Pipe flow

$$Nu_D = \frac{(f/8)(Re_D - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \Rightarrow 3000 \leq Re_D \leq 5 \times 10^6$$

Turbulent Heat Transfer over a Flat Plate

For Laminar flow, we derived:

$$C_{f,x} = \frac{\bar{\tau}_{0,x}}{\frac{1}{2} \rho V_{\infty}^2} = \frac{0.664}{Re_x^{1/2}}$$

$$C_{f,av} = \frac{1.328}{Re_L^{1/2}}$$

$$\frac{\delta}{x} = \frac{4.92}{Re_x^{1/2}}$$

$$Nu_x = 0.332 Re_x^{1/2} Pr^{1/3} \Rightarrow \overline{Nu}_L = \frac{\bar{h} L}{k} = 0.664 Re_L^{1/2} Pr^{1/3}$$

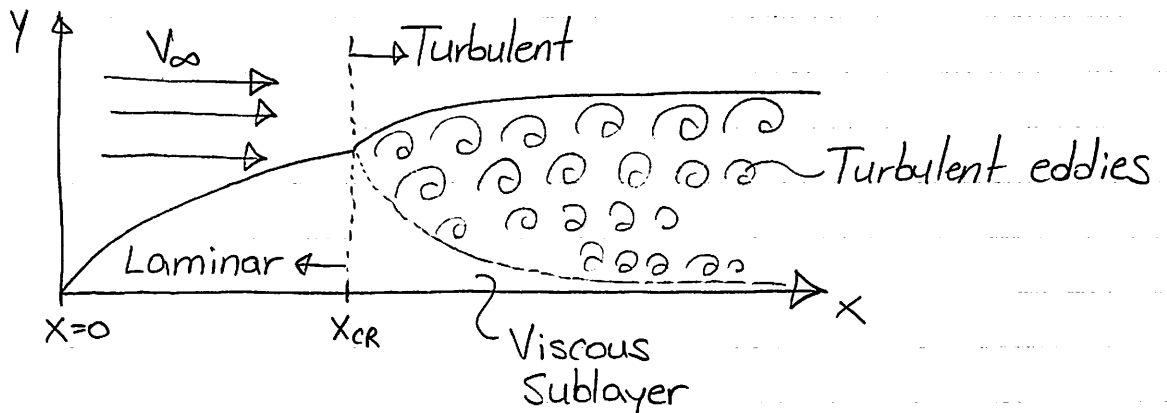
$$St_x \cdot Pr^{2/3} = \frac{C_{f,x}}{2}$$

$C_{f,x}$ = skin friction coefficient at location x along plate

$\bar{\tau}_{0,x}$ = shear stress at plate surface at location x along plate

$C_{f,av}$ = average friction coefficient over the whole plate.

Now we can examine the turbulent flow regime



$$Re_{CR} = \frac{V_{\infty} x_{CR}}{\nu} = 50,000 - 500,000$$

Due to the difficulty in analytically modeling the turbulent boundary layer on a flat plate, many of the following expressions stem from experimental correlations.

$$C_{f,x} = 0.0592 Re_x^{-1/5}$$

$$C_{f,x} = 0.026 Re_x^{-1/7}$$

$$10^5 \leq Re_x \leq 10^7$$

$$10^7 \leq Re_x \leq 10^9$$

$\Rightarrow$ Turbulent flow skin friction coefficient.

For even greater accuracy: (Developed by White)

$$\boxed{C_{f,x} = \frac{0.455}{(\ln 0.06 Re_x)^2} ; 10^5 \leq Re_x \leq 10^9} \Rightarrow \text{Turbulent flow over a flat plate.}$$

Note, x is still the distance from the beginning of the flat plate, not from x_{CR} .

Typically, we want the average skin friction coefficient for the whole plate

$$\begin{aligned} \vec{\tau}_{0,x} &= \frac{1}{2} \rho V_\infty^2 C_{f,x} = \frac{1}{2} \rho V_\infty^2 \cdot 0.0592 Re_x^{-0.2} \\ &= \frac{1}{2} \rho V_\infty^2 \cdot 0.0592 \left(\frac{\rho V_\infty x}{\mu} \right)^{-0.2} \\ &= C x^{-0.2} \quad \text{where } C = \frac{1}{2} \rho V_\infty^2 \cdot 0.0592 \left(\frac{\rho V_\infty}{\mu} \right)^{-0.2} \end{aligned}$$

$$\vec{\tau}_{av,L} = \frac{C}{L} \int_0^L \frac{dx}{x^{0.2}} = \frac{C x^{0.8}}{0.8L} = \frac{C}{0.8L^{0.2}} = \frac{CL^{-0.2}}{0.8} = \frac{\vec{\tau}_{0,L}}{0.8}$$

We can now solve for $C_{f,av}$

$$C_{f,av} = \frac{C_{f,L}}{0.8} = \frac{0.0592 Re_L^{-1/5}}{0.8}$$

$$\boxed{C_{f,av} = 0.074 Re_L^{-1/5}}$$

Applying our Colburn analogy: $St_x = \frac{Nu_x}{Re_x \cdot Pr}$ (Stanton number)

We know for a flat plate: $St_x Pr^{2/3} = \frac{C_{f,x}}{2}$

$$\frac{Nu_x}{Re_x Pr} \cdot Pr^{2/3} = \frac{C_{f,x}}{2}$$

$$Nu_x = \frac{C_{f,x}}{2} Re_x Pr^{1/3}$$

Using $C_{f,x} = 0.0592 Re_x^{-1/5}$

$$Nu_x = \frac{0.0592}{2} Re_x^{-1/5} \cdot Re_x Pr^{1/3}$$

$$Nu_x = 0.029 Re_x^{0.8} Pr^{1/3} \Rightarrow \text{Colburn correlation (from analogy)}$$

$$Nu_x = 0.029 Re_x^{0.8} Pr^{0.43} \Rightarrow \text{Whitaker correlation}$$

$$\overline{Nu_L} = \frac{\bar{h}L}{k_f} = 0.036 Re_L^{0.8} Pr^{0.43} \Rightarrow \begin{cases} 0.7 < Pr < 400 \\ 5 \times 10^5 < Re_x < 3 \times 10^7 \end{cases}$$

For a more accurate correlation

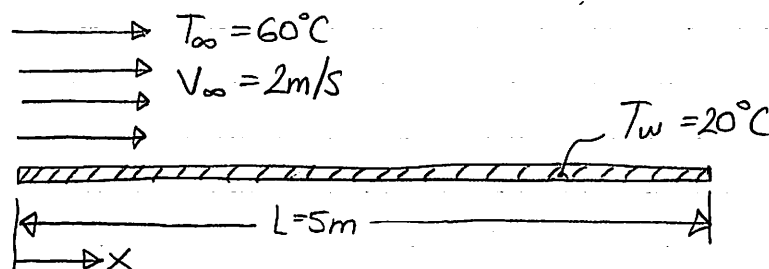
$$Nu_x = \frac{\left(\frac{C_{f,x}}{2}\right) \cdot Re_x Pr}{1 + 12.7 \left(\frac{C_{f,x}}{x}\right)^{1/2} (Pr^{2/3} - 1)} \Rightarrow \text{White correlation}$$

$0.5 < Pr < 2000$
 $5 \times 10^5 < Re_x < 10^7$

Note, the closest we got to an analytical result was with the Colburn analogy, but even then we had to use the local skin friction coefficient ($C_{f,x}$) based on experiments.

For more reading, see page 312 in Mills textbook.

Example Engine oil at 60°C flows over a 5m long flat plate whose temperature is 20°C with a velocity of 2 m/s. Determine the total drag force and heat transfer per unit width of the entire plate.



Assume : 1) $T_w = \text{constant} \neq f(x) = 20^\circ\text{C}$
 2) Constant properties, steady flow.

$$T_f = \bar{T} = \frac{T_w + T_\infty}{2} = 40^\circ\text{C}$$

$$\rho_{oil} = 876 \text{ kg/m}^3$$

$$k_{oil} = 0.144 \text{ W/m}\cdot\text{K}$$

$$Pr = 2870 \text{ (Note how large this Prandtl number is!)}$$

$$\nu_{oil} = 242 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Pr_{water} \approx 7$$

$$Pr_{air} \approx 1$$

First we check if we are in the turbulent or laminar regime

$$Re_L = \frac{V_\infty L}{\nu_{oil}} = 4.13 \times 10^4 < 5.0 \times 10^5 \text{ (Laminar flow)}$$

We know from our previous derivations (page 85 of notes)

$$C_{f,av} = \frac{\bar{C}_{o,av}}{\frac{1}{2} \rho V_\infty^2} = 1.328 Re_L^{-0.5} = 0.00653$$

$$F_D = C_{f,av} \cdot A \cdot \frac{\rho V_\infty^2}{2} = (0.00653)(5\text{m} \cdot 1\text{m}) \frac{(876 \text{ kg/m}^3)(2\text{m/s})^2}{2}$$

$$\boxed{F_D = 57.2 \text{ N}}$$

For heat transfer:

$$Nu_L = \frac{\bar{h} L}{k_{oil}} = 0.664 Re_L^{1/2} Pr^{1/3} = 1918$$

$$\bar{h} = 55.2 \text{ W/m}^2 \cdot \text{K}$$

$$\boxed{Q = hA(T_\infty - T_w) = 11.04 \text{ kW}}$$

Now what if we 15x the speed of our flow? Will our friction and heat transfer 15x as well?

$$Re_L = \frac{V_\infty L}{\nu_{oil}} = 6.2 \times 10^5 > Re_{crit} \text{ (Turbulent flow)}$$

$$C_{f,av} = 0.074 Re_L^{-0.2} = 0.00514 \text{ (page 153 of notes)}$$

$$F_D = C_{f,av} \cdot A \cdot \frac{\rho V_\infty^2}{2} = (0.00514)(5m^2) \frac{(876 kg/m^3)(90m/s)^2}{2}$$

$$F_D = 10.13 kN$$

For heat transfer:

$$\overline{Nu}_L = \frac{\overline{h} L}{k_{oil}} = 0.036 Re_L^{0.8} Pr^{0.43} \text{ (Estimate only since } Pr > 400)$$

$$= 0.036 (6.2 \times 10^5)^{0.8} (2870)^{0.43} = 47545 \quad \begin{matrix} \leftarrow \text{Could have used} \\ \text{White correlation} \end{matrix}$$

$$\overline{h} = \frac{\overline{Nu}_L \cdot k_{oil}}{L} = \frac{(47545)(0.144 W/m \cdot K)}{5m}$$

$$\overline{h} = 1369.3 W/m^2 \cdot K$$

$$Q = \overline{h} A (T_\infty - T_w) = 273.9 kW$$

Now let's compare:

$$\frac{V_{\infty,2}}{V_{\infty,1}} = \frac{15}{1} = 15$$

$$\frac{F_{D,2}}{F_{D,1}} = \frac{10130N}{57.2N} = 177.1 \text{ (Drag increases dramatically)}$$

$$\frac{Q_2}{Q_1} = \frac{273.9 kW}{11.04 kW} = 24.8 \text{ (Heat transfer increases modestly)}$$

Note, this is where you strike a balance in heat exchanger design in terms of heat transfer and pressure drop.

Note also, another important factor to keep in mind is the working fluid. In this case, oil is used with a high Prandtl number and high viscosity. If we had chosen water or air, how would our results change:

Water: $\rho_w = 1000 \text{ kg/m}^3$
 $k_w = 0.6 \text{ W/m}\cdot\text{K}$
 $Pr = 7$
 $\nu_w = 0.658 \times 10^{-6} \text{ m}^2/\text{s}$

$$Re_L = \frac{V_\infty L}{\nu_w} = \frac{(2 \text{ m/s})(5 \text{ m})}{0.658 \times 10^{-6} \text{ m}^2/\text{s}} = 1.52 \times 10^7 \text{ (Turbulent flow)}$$

$$C_{f,av} = 0.074 Re_L^{-0.2} = 0.0027$$

$$F_{D,w} = C_{f,av} \cdot A \cdot \frac{\rho V_\infty^2}{2} = (0.0027)(5 \text{ m}^2) \frac{(1000 \text{ kg/m}^3)(2 \text{ m/s})^2}{2}$$

$$F_{D,w} = 27.1 \text{ N}$$

For heat transfer:

$$\overline{Nu}_L = \frac{\bar{h} L}{k_w} = 0.036 Re_L^{0.8} Pr^{0.43}$$

$$= 0.036 (1.52 \times 10^7)^{0.8} (7)^{0.43} = 46256.2$$

$$\bar{h} = \frac{\overline{Nu}_L \cdot k_w}{L} = \frac{(46256.2)(0.6 \text{ W/m}\cdot\text{K})}{5 \text{ m}}$$

$$\bar{h} = 5550.8 \text{ W/m}^2 \cdot \text{K}$$

$$Q_w = \bar{h} A (T_\infty - T_w) = 1.11 \text{ MW}$$

$$\frac{F_{D,oil}}{F_{D,w}} = 2.11 \quad \frac{Q_{oil}}{Q_w} = 0.01 \text{ or } 1\% \text{ only.}$$

This simple analysis shows the importance of selecting the proper working fluid. You may be asking yourself, why would anybody use a high Pr oil as a working fluid in a thermal application? Isn't water always better?

The answer is NO! Oils have 2 main advantages over water.

- 1) Temperature range (Typically -40 to 300°C). Water will freeze or boil unless its pressurized which makes the advantages of water not worth it (pressurization is tricky and requires a lot of design, maintenance, and for water, high pressures).
- 2) Low volatility. Water has a relatively high vapor pressure meaning it evaporates and coolant can be lost and needs to be replaced. Oils vapor pressure at $T = 0-60^{\circ}\text{C}$ is usually 0 Pa , meaning no evaporative mass loss.
- 3) Corrosion resistance and enhanced lubrication.

Cylinders and Spheres

So far, we've only looked at internal flow in a tube, and external flow on a flat plate. Most heat exchangers are not in this configuration.

For example: Shell and tube condenser

